Lesson 2: Baseline Email Assistant

This lesson builds an email assistant that:

- Classifies incoming messages (respond, ignore, notify)
- Drafts responses
- Schedules meetings

We'll start with a simple implementation - one that uses hard-coded rules to handle emails.

> 💻 **Access `requirements.txt` , notebooks and other files:** 1) click on the *"File"* option on the top menu of the notebook and then 2) click on *"Open"*.
>
> ⬇️ **Download Notebooks:** 1) click on the *"File"* option on the top menu of the notebook and then 2) click on *"Download as"* and select *"Notebook (.ipynb)"*.
>
> 📒 For more help, please see the *"Appendix – Tips, Help, and Download"* Lesson.

> 🚨 **Different Run Results:** The output generated by AI chat models can vary with each execution due to their dynamic, probabilistic nature. Don't be surprised if your results differ from those shown in the video.

Load API tokens for our 3rd party APIs

In [1]:
```
import os
from dotenv import load_dotenv
_ = load_dotenv()
```

Setup a Profile, Prompt and Example Email

In [2]:
```
profile = {
    "name": "John",
    "full_name": "John Doe",
    "user_profile_background": "Senior software engineer leading a team of 5
}
```

In [3]:
```
prompt_instructions = {
    "triage_rules": {
        "ignore": "Marketing newsletters, spam emails, mass company announce
        "notify": "Team member out sick, build system notifications, project
        "respond": "Direct questions from team members, meeting requests, cr
    },
    "agent_instructions": "Use these tools when appropriate to help manage J
}
```

In [15]:
```
# Example incoming email
email = {
    "from": "Alice Smith <alice.smith@company.com>",
    "to": "John Doe <john.doe@company.com>",
    "subject": "🎉 Congratulations! You've Won $1,500,000 in the Global Lott
    "body": """
Hi John,We are pleased to inform you that your email address was randomly se

Winning Amount: $1,500,000 USD
Reference Number: GML/0425/INTL

This draw was conducted by an automated email ballot system in collaboratior

To claim your prize, kindly provide the following details within the next 72

Full Name

Age & Country of Residence

Mobile Number

Occupation

Please reply to this email or contact our claims agent:
📧 Mr. James Peterson – claims@gmlotteryclaims.com

Important: Failure to respond within 3 days may lead to disqualification and

Specifically, I'm looking at:
- /auth/refresh
- /auth/validate

Thanks!
Alice""",
}
```

Define the first part of the agent - triage.

In [16]:
```
from pydantic import BaseModel, Field
from typing_extensions import TypedDict, Literal, Annotated
from langchain.chat_models import init_chat_model
```

In [17]:
```
llm = init_chat_model("openai:gpt-4o-mini")
```

In [18]:
```python
class Router(BaseModel):
    """Analyze the unread email and route it according to its content."""

    reasoning: str = Field(
        description="Step-by-step reasoning behind the classification."
    )
    classification: Literal["ignore", "respond", "notify"] = Field(
        description="The classification of an email: 'ignore' for irrelevant
        "'notify' for important information that doesn't need a response, "
        "'respond' for emails that need a reply",
    )
```

In [19]:
```python
llm_router = llm.with_structured_output(Router)
```

In [20]:
```python
from prompts import triage_system_prompt, triage_user_prompt
```

In [21]:
```python
# uncomment to view
print(triage_system_prompt)
print(triage_user_prompt)
```

```
< Role >
You are {full_name}'s executive assistant. You are a top-notch executive ass
istant who cares about {name} performing as well as possible.
</ Role >

< Background >
{user_profile_background}.
</ Background >

< Instructions >

{name} gets lots of emails. Your job is to categorize each email into one of
three categories:

1. IGNORE - Emails that are not worth responding to or tracking
2. NOTIFY - Important information that {name} should know about but doesn't
require a response
3. RESPOND - Emails that need a direct response from {name}

Classify the below email into one of these categories.

</ Instructions >

< Rules >
Emails that are not worth responding to:
{triage_no}

There are also other things that {name} should know about, but don't require
an email response. For these, you should notify {name} (using the `notify` r
esponse). Examples of this include:
{triage_notify}

Emails that are worth responding to:
{triage_email}
</ Rules >

< Few shot examples >
{examples}
</ Few shot examples >


Please determine how to handle the below email thread:

From: {author}
To: {to}
Subject: {subject}
{email_thread}
```

In [22]:
```python
system_prompt = triage_system_prompt.format(
    full_name=profile["full_name"],
    name=profile["name"],
    examples=None,
    user_profile_background=profile["user_profile_background"],
    triage_no=prompt_instructions["triage_rules"]["ignore"],
    triage_notify=prompt_instructions["triage_rules"]["notify"],
```

```
    triage_email=prompt_instructions["triage_rules"]["respond"],
)
```

In [23]:
```
user_prompt = triage_user_prompt.format(
    author=email["from"],
    to=email["to"],
    subject=email["subject"],
    email_thread=email["body"],
)
```

In [24]:
```
result = llm_router.invoke(
    [
        {"role": "system", "content": system_prompt},
        {"role": "user", "content": user_prompt},
    ]
)
```

In [25]:
```
print(result)
```

```
reasoning='This email is a classic lottery scam. It claims that John has won
a large sum of money and requests personal information to claim the prize. S
uch emails are typically spam and should not be taken seriously. Therefore,
this email does not warrant a response or further action.' classification='i
gnore'
```

Main agent, define tools

In [27]:
```
from langchain_core.tools import tool
```

In [28]:
```
@tool
def write_email(to: str, subject: str, content: str) -> str:
    """Write and send an email."""
    # Placeholder response - in real app would send email
    return f"Email sent to {to} with subject '{subject}'"
```

In [29]:
```
@tool
def schedule_meeting(
    attendees: list[str],
    subject: str,
    duration_minutes: int,
    preferred_day: str
) -> str:
    """Schedule a calendar meeting."""
    # Placeholder response - in real app would check calendar and schedule
    return f"Meeting '{subject}' scheduled for {preferred_day} with {len(att
```

In [30]:
```
@tool
def check_calendar_availability(day: str) -> str:
    """Check calendar availability for a given day."""
    # Placeholder response - in real app would check actual calendar
    return f"Available times on {day}: 9:00 AM, 2:00 PM, 4:00 PM"
```

Main agent, define prompt

In [31]:
```python
from prompts import agent_system_prompt
def create_prompt(state):
    return [
        {
            "role": "system",
            "content": agent_system_prompt.format(
                instructions=prompt_instructions["agent_instructions"],
                **profile
                )
        }
    ] + state['messages']
```

In [32]:
```python
print(agent_system_prompt)
```

```
< Role >
You are {full_name}'s executive assistant. You are a top-notch executive ass
istant who cares about {name} performing as well as possible.
</ Role >

< Tools >
You have access to the following tools to help manage {name}'s communication
s and schedule:

1. write_email(to, subject, content) - Send emails to specified recipients
2. schedule_meeting(attendees, subject, duration_minutes, preferred_day) - S
chedule calendar meetings
3. check_calendar_availability(day) - Check available time slots for a given
day
</ Tools >

< Instructions >
{instructions}
</ Instructions >
```

In [33]:
```python
from langgraph.prebuilt import create_react_agent
```

In [34]:
```python
tools=[write_email, schedule_meeting, check_calendar_availability]
```

In [35]:
```python
agent = create_react_agent(
    "openai:gpt-4o",
    tools=tools,
    prompt=create_prompt,
)
```

In [42]:
```
response = agent.invoke(
    {"messages": [{
        "role": "user",
        "content": "what is my availability for today?"
    }]}
)
```

In [43]:
```
response["messages"][-1].pretty_print()
```

```
================================== Ai Message ==================================

You have available time slots today at 9:00 AM, 2:00 PM, and 4:00 PM.
```

Create the Overall Agent

In [44]:
```
from langgraph.graph import add_messages

class State(TypedDict):
    email_input: dict
    messages: Annotated[list, add_messages]
```

In [45]:
```
from langgraph.graph import StateGraph, START, END
from langgraph.types import Command
from typing import Literal
from IPython.display import Image, display
```

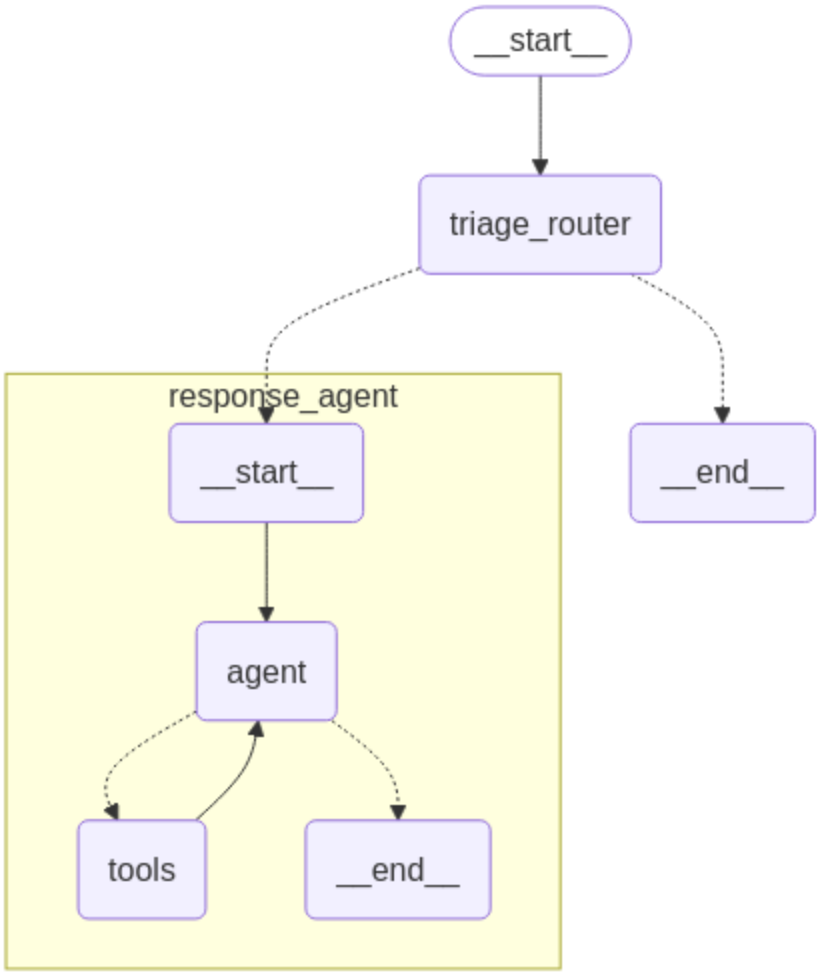
In [46]:
```
def triage_router(state: State) -> Command[
    Literal["response_agent", "__end__"]
]:
    author = state['email_input']['author']
    to = state['email_input']['to']
    subject = state['email_input']['subject']
    email_thread = state['email_input']['email_thread']

    system_prompt = triage_system_prompt.format(
        full_name=profile["full_name"],
        name=profile["name"],
        user_profile_background=profile["user_profile_background"],
        triage_no=prompt_instructions["triage_rules"]["ignore"],
        triage_notify=prompt_instructions["triage_rules"]["notify"],
        triage_email=prompt_instructions["triage_rules"]["respond"],
        examples=None
    )
    user_prompt = triage_user_prompt.format(
        author=author,
        to=to,
        subject=subject,
        email_thread=email_thread
    )
    result = llm_router.invoke(
        [
            {"role": "system", "content": system_prompt},
```

```
            {"role": "user", "content": user_prompt},
        ]
    )
    if result.classification == "respond":
        print("📧 Classification: RESPOND - This email requires a response")
        goto = "response_agent"
        update = {
            "messages": [
                {
                    "role": "user",
                    "content": f"Respond to the email {state['email_input']}
                }
            ]
        }
    elif result.classification == "ignore":
        print("🚫 Classification: IGNORE - This email can be safely ignored"
        update = None
        goto = END
    elif result.classification == "notify":
        # If real life, this would do something else
        print("🔔 Classification: NOTIFY - This email contains important inf
        update = None
        goto = END
    else:
        raise ValueError(f"Invalid classification: {result.classification}")
    return Command(goto=goto, update=update)
```

Put it all together

In [47]:
```
email_agent = StateGraph(State)
email_agent = email_agent.add_node(triage_router)
email_agent = email_agent.add_node("response_agent", agent)
email_agent = email_agent.add_edge(START, "triage_router")
email_agent = email_agent.compile()
```

In [48]:
```
# Show the agent
display(Image(email_agent.get_graph(xray=True).draw_mermaid_png()))
```



In [49]:
```
email_input = {
    "author": "Marketing Team <marketing@amazingdeals.com>",
    "to": "John Doe <john.doe@company.com>",
    "subject": "🔥 EXCLUSIVE OFFER: Limited Time Discount on Developer Tools
    "email_thread": """Dear Valued Developer,

Don't miss out on this INCREDIBLE opportunity!

🚀 For a LIMITED TIME ONLY, get 80% OFF on our Premium Developer Suite!

✨ FEATURES:
- Revolutionary AI-powered code completion
- Cloud-based development environment
- 24/7 customer support
- And much more!

💰 Regular Price: $999/month
🎉 YOUR SPECIAL PRICE: Just $199/month!

🕒 Hurry! This offer expires in:
24 HOURS ONLY!

Click here to claim your discount: https://amazingdeals.com/special-offer

Best regards,
Marketing Team
```

```
---
To unsubscribe, click here
""",
}
```

In [50]:
```
response = email_agent.invoke({"email_input": email_input})
```

```
🚫 Classification: IGNORE - This email can be safely ignored
```

In [51]:
```
email_input = {
    "author": "Alice Smith <alice.smith@company.com>",
    "to": "John Doe <john.doe@company.com>",
    "subject": "Quick question about API documentation",
    "email_thread": """Hi John,

I was reviewing the API documentation for the new authentication service and

Specifically, I'm looking at:
- /auth/refresh
- /auth/validate

Thanks!
Alice""",
}
```

In [52]:
```
response = email_agent.invoke({"email_input": email_input})
```

```
📧 Classification: RESPOND - This email requires a response
```

In [53]:
```
for m in response["messages"]:
    m.pretty_print()
```

```
================================ Human Message =================================

Respond to the email {'author': 'Alice Smith <alice.smith@company.com>', 'to': 'John Doe <john.doe@company.com>', 'subject': 'Quick question about API documentation', 'email_thread': "Hi John,\n\nI was reviewing the API documentation for the new authentication service and noticed a few endpoints seem to be missing from the specs. Could you help clarify if this was intentional or if we should update the docs?\n\nSpecifically, I'm looking at:\n- /auth/refresh\n- /auth/validate\n\nThanks!\nAlice"}
================================== Ai Message ==================================
Tool Calls:
  write_email (call_1ndxzPIinVSvav1pBioakpQR)
 Call ID: call_1ndxzPIinVSvav1pBioakpQR
  Args:
    to: alice.smith@company.com
    subject: Re: Quick question about API documentation
    content: Hi Alice,

Thank you for reaching out with your question about the API documentation. I can confirm that the endpoints /auth/refresh and /auth/validate should indeed be included in the documentation. It was an oversight, and the documentation needs to be updated.

I'll ensure that the team updates the documentation to include these endpoints as soon as possible.

Best regards,

John Doe
================================= Tool Message =================================
Name: write_email

Email sent to alice.smith@company.com with subject 'Re: Quick question about API documentation'
================================== Ai Message ==================================

I have responded to Alice's email, clarifying that the endpoints she mentioned should be included in the documentation and that the team will update the documentation accordingly.
```

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